

WASH_Mortality_Data_Science_ Project_Report

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Links

Dataset : https://data360.worldbank.org/en/indicator/WB_WDI_SH_STA_WASH_P5

Link to Streamlit app:

<https://wash-mortality-dashboard-49okovwszkkndcducpw9ns.streamlit.app/>

1. Aims and Objectives

The dataset utilized in this project is the Mortality Rate for children under 5, Caused by Unsafe Water, Sanitation, and Hygiene (WASH) Practices (% of deaths in under-5s), 2019, obtained from the World Bank Data360 platform

https://data360.worldbank.org/en/indicator/WB_WDI_SH_STA_WASH_P5

After cleaning, the data set comprises 176 entries (rows) and 4 features (columns): Country Code, Country Name, Mortality Rate 2019, and Risk Category. This dataset was chosen from the World Bank, which is among the authorized data suppliers mentioned in the coursework guidelines

1.1 Aims

The aim of this project is to create an interactive data dashboard that examines and displays the worldwide under-5 child mortality rate linked to poor WASH (Water, Sanitation, and Hygiene) conditions in 2019. The dashboard aims to assist senior executives, financial specialists, and tech professionals at an international sustainability conference by delivering valuable data insights in an engaging and user-friendly manner

1.2 Objectives

- i. To illustrate the worldwide geographic spread of WASH related under 5 mortality rates with an interactive choropleth map of the world.
- ii. To allow users to actively filter and examine the data based on risk category and range of mortality rates.
- iii. To determine and classify the countries with the highest risk using an interactive bar chart.
- iv. To display the statistical distribution and variability of mortality rates among risk categories through histograms and box plots.
- v. To offer a searchable data table enabling users to find specific countries.
- vi. To emphasize three essential findings that back evidence informed decision making related to WASH investment priorities

1.3 Justification for the selected Key Insights

The dashboard emphasized three primary insights. Initially, the identification of the countries with the highest mortality rates : Lesotho (108.1 per 1,000 live births), Chad (99.2), and Somalia (99.2) was chosen as these severe instances are readily

actionable for conference delegates tasked with managing aid or investment. Additionally, the observation that 7.95% of nations are classified in the "Very High" risk category while accounting for an unequal portion of child fatalities was chosen to convey the severity of the issue and the need for focused intervention. Third, the statistic indicating that 43.2% of countries are in the "Low" risk category was included to offer a balanced viewpoint, highlighting global advancements while emphasizing the inequality gap. These three insights together present a thorough and engaging narrative for an audience focused on sustainability

2. Development Methodology

2.1 Selected Approach: Iterative (Agile-Inspired) Development

A development approach inspired by agile and based on iteration was used for this project. In this method, the dashboard was developed gradually adding one feature or component at a time, with testing and evaluation performed after each iteration before moving on to the next. This is different from a linear Waterfall method, where all planning and design must be finished prior to starting any development

Justification for the selected approach:

The iterative method was selected for various reasons. Initially, the brief project timeline (around five weeks) required a flexible, quick-paced approach instead of an extensive preliminary design stage. Secondly, Streamlit's design is fundamentally iterative, modifications to the Python source file are instantly visible in the browser, allowing for quick and efficient prototyping. Third, the particular visualizations and interactive features were not entirely established at the beginning; certain necessities (like the sort order dropdown and the show/hide toggle buttons) surfaced during development as enhancements for usability. A repetitive approach adapted to these changing needs without interrupting the overall development workflow. A Waterfall approach would have been less appropriate since it relies on stable requirements, which was not the situation in this instance

2.2 Key Phases of Development:

Phase 1 - Data Selection and Preparation:

The dataset was obtained from the Data360 portal of the World Bank. The raw data was reviewed and cleaned: unnecessary columns were eliminated, the dataset was

condensed to four essential columns (Country_Code, Country_Name, Mortality_Rate_2019, Risk_Category), and rows with missing values were handled. The dataset was cleaned, saved as a CSV file, and uploaded to the GitHub repository.

Phase 2 - Setup of Environment and Repository :

A public repository on GitHub called wash-mortality-dashboard was established. A Python development setup was established with the required libraries: Streamlit, Pandas, and Plotly. A requirements.txt file was generated to guarantee consistency during deployment. A project structure was established with the first commit

Phase 3 -Core Dashboard Development:

The Streamlit app was created in an iterative manner. The design included a sidebar for user filters and a primary content section for visualizations. Components were developed and integrated in this order: page setup and data retrieval, KPI metrics row, choropleth world map, donut chart, sortable bar chart, histogram and box plot, key insights area, and a data table with search functionality. Every component was pushed to GitHub with a detailed commit message.

Phase 4 -Testing and Refinement:

All functional requirements were evaluated through established test cases. While testing, an unhandled error was found when all risk categories were unchecked ,this was fixed by incorporating input validation with st.error() and st.stop(). A second issue was found in which the bar chart did not accurately represent the sort order dropdown,this was resolved by modifying the ascending parameter according to the dropdown value. The test log was finalized with real results and pass/fail statistics.

Phase 5 – Deployment

The completed application was launched on Streamlit Community Cloud by linking the public GitHub repository. The deployment was validated by checking the live public URL across various browsers and ensuring all interactive components worked properly

3. Requirements

Requirements are categorized into Functional (FR), outlining what the system needs to accomplish or Non-Functional (NFR) ,detailing how the system should operate.

3.1 Functional Requirements

ID	Requirement name	Requirement description
FR1	Data Visualization	Show a World Map, Bar Chart, and Histogram for mortality data.
FR2	Multi-Category Filter	Sort data by Risk Categories (Low, Moderate, High, Very High)
FR3	Range Slider	Select countries according to a specified range of numeric Mortality Rates
FR4	Search Box	Narrow down the data table to a particular country by its name
FR5	Dynamic KPIs	Update summary stats (Highest, Lowest, Average) in real-time.
FR6	Top-N Control	Change the quantity of bars displayed in the Bar Chart (for instance, Top 5, Top 25)

3.2 Non-Functional Requirements

ID	Requirement name	Goal
NFR1	Performance	Every filter needs to refresh visuals in less than 2 seconds

NFR2	Usability	The interface should be user-friendly for individuals lacking technical background
NFR3	Accessibility	Available through regular web browsers (Chrome/Edge) using Streamlit
NFR4	Accuracy	KPI calculations need to remain fully consistent with the filtered data

4. Test Cases

#	TC1	Title:	Loading Data and Verifying KPIs
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Description	Ensure the application correctly loads the dataset upon startup and shows precise KPI values
Steps and input data	1. Launch the application at localhost:8501. 2. Make sure all sidebar filters are set to default (all categories chosen, complete rate range, N=15) 3. Observe the four KPI metric cards
Dependencies	The file cleaned_mortality_data.csv needs to be located in the project folder
Expected result	KPI cards exhibit: Countries Represented = 176, Maximum Rate = 108.10, Minimum Rate = 0.40, Mean Rate = 16.82. No error is shown

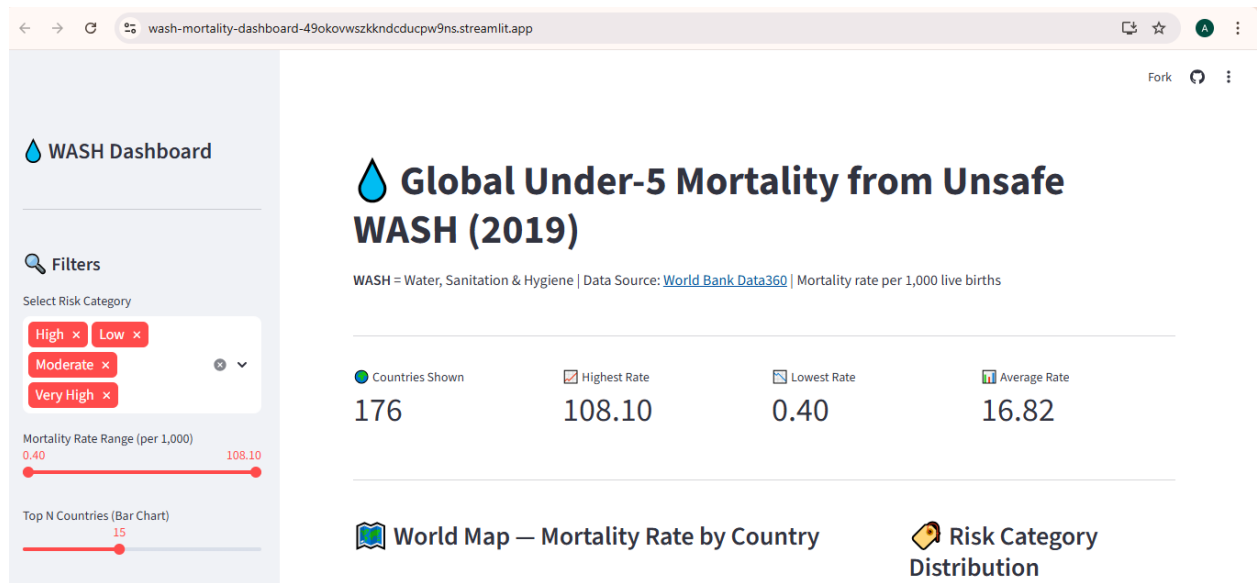


Figure i : Dashboard initial state showing default KPIs for 176 countries

#	TC2	Title:	Loading Data and Verifying KPIs
Description	Ensure that choosing one risk category accurately filters all graphs and the data table		

Steps and input data	1. In the sidebar, uncheck "Low", "Moderate" and "High" 2. Retain only the "Very High" option selected 3. Examine the map of the world, the bar graph, and the data table
Dependencies	TC1 must pass
Expected result	All visualizations refresh to display solely Very High risk countries. KPI "Countries Shown" decreases to 14. No error is shown

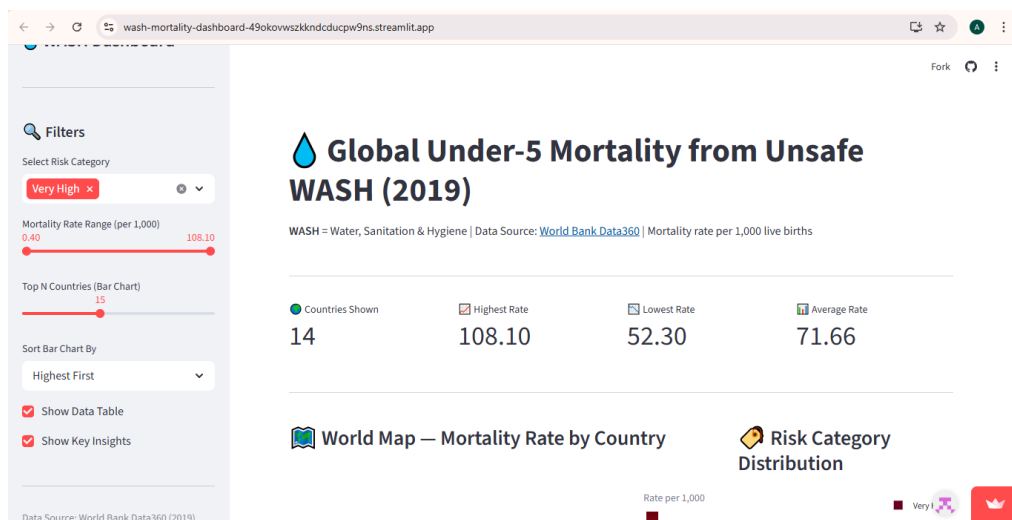


Figure ii: Multi-select filter applied for 'Very High' risk category

	Country_Code	Country_Name	Mortality_Rate_2019	Risk_Category
1	LSO	Lesotho	108.1	Very High
2	TCD	Chad	99.2	Very High
3	SOM	Somalia, Federal Republic of	99.2	Very High
4	NGA	Nigeria	71.7	Very High
5	NER	Niger	70.3	Very High
6	SLE	Sierra Leone	69.5	Very High
7	SSD	South Sudan	68.1	Very High
8	ERI	Eritrea	66.5	Very High
9	MLI	Mali	66.1	Very High
10	BFA	Burkina Faso	60.9	Very High
11	BEN	Benin	60.2	Very High
12	GIN	Guinea	57.8	Very High
13	BDI	Burundi	53.3	Very High
14	COD	Congo, Dem. Rep.	52.3	Very High

Figure iii: updating KPIs to 14 countries.

#	TC3	Title:	Slider for Mortality Rate Range
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Description	Ensure that modifying the mortality rate slider properly updates the data throughout all sections of the dashboard
Steps and input data	1. Restore all filters to their default settings 2. Shift the left handle of the mortality slider to 50.30 3. Observe the world map, bar chart, and data table
Dependencies	TC1 must pass
Expected result	Displayed are only countries with a mortality rate of 50.30 or higher. The data table contains a reduced number of rows. KPI values are refreshed to show filtered data

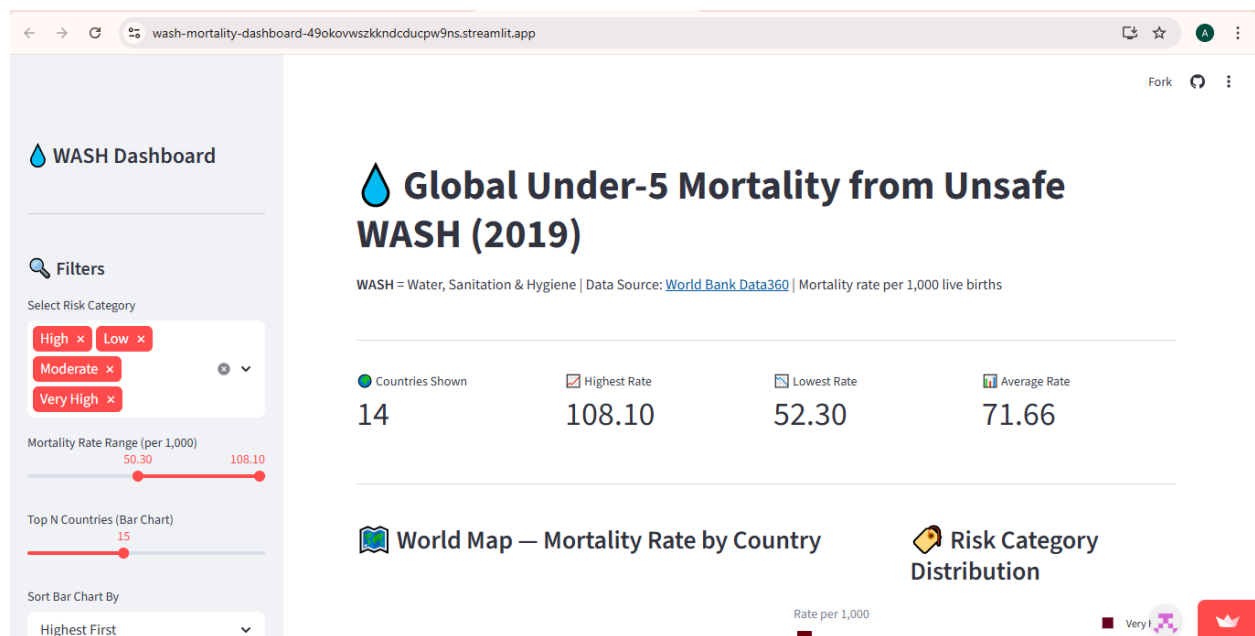


Figure iv: Mortality rate range slider set to 50.30

#	TC4	Title:	Slider for Top N Bar Chart
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Description	Ensure that the Top N Countries slider accurately regulates the quantity of bars displayed in the bar chart
Steps and input data	1. Restore all filters to default settings. 2. Adjust the Top N slider to 5 3. Count the bars in the fourth bar chart. 4. Adjust the slider to 25 5. Recount the bars once more
Dependencies	TC1 must pass
Expected result	The bar chart displays precisely 5 bars for N=5 and exactly 25 bars for N=25. Displayed countries represent the accurate top N based on mortality rate



Figure vi: Bar chart dynamically scaled to display the Top 5 countries



Figure v: Bar chart expanded to display the Top 25 countries

#	TC5	Title:	Search for Country Name in Data Table
Description	Ensure that entering a country name in the search field accurately narrows down the data table to display solely the corresponding rows		
Steps and input data	1. Return all filters to their original default settings. 2. Confirm that the "Show Data Table" option is checked 3. Enter "Nigeria" into the search field		
Dependencies	TC1 must pass. Show Data Table checkbox must be enabled.		
Expected result	Data table shows exactly one row: NGA, Nigeria, 71.7, Very High. Info message updates to "Showing 1 countries."		

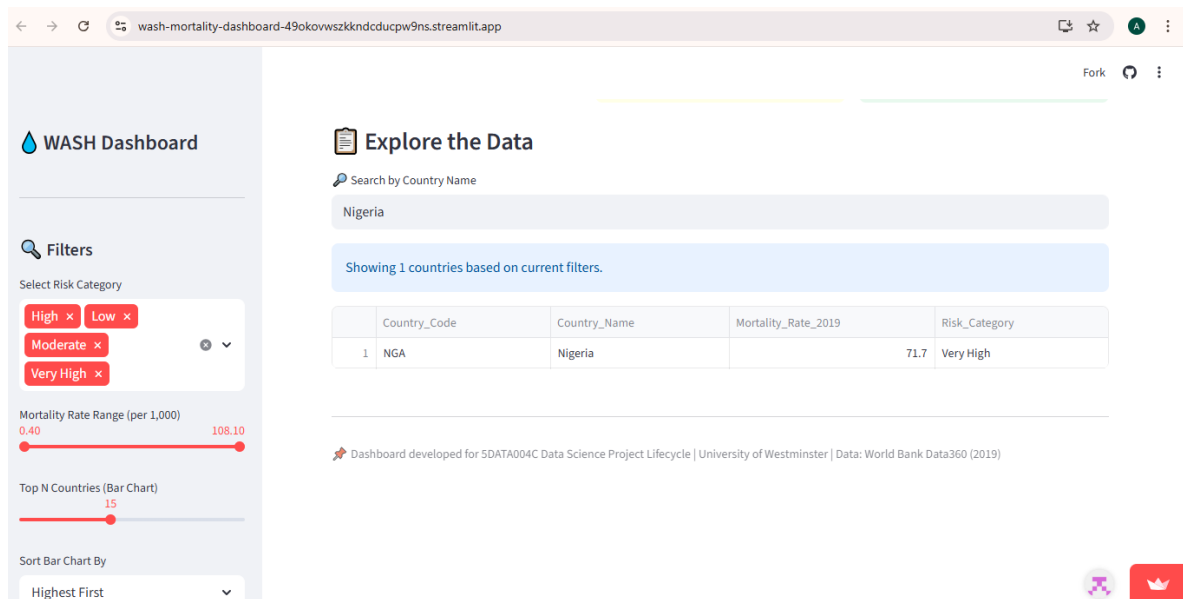


Figure vii: Keyword search for "Nigeria" returning a single matching record in the data table

5. Test log

TC	Date	Executed by	Actual result	Pass/Fail	Notes
TC1	23/04/2026	AzmaLahir	KPI showed 176 nations, Highest 108.10, Lowest 0.40, Average 16.82. App launched in less than 2 seconds	Pass	No issues found
TC2	23/04/2026	AzmaLahir	All charts refreshed to Very High exclusively. Countries displayed decreased to 14. The data table displayed solely Very High entries	Pass	No issues found
TC3	23/04/2026	AzmaLahir	Displayed were only countries with a rate of 50.30 or higher. The data table and KPIs were accurately updated to reflect 14 countries	Pass	No issues found
TC4	23/04/2026	AzmaLahir	The bar chart showed exactly 5 bars when N=5 and 25 bars when N=25. Accurate nations displayed in both instances	Pass	No issues found
TC5	23/04/2026	23/04/2026	Data table adjusted to display a single row: NGA, Nigeria, 71.7, Very High. Info message revised to "Displaying 1 country"	Pass	Minor grammatical error in the information message ("1 countries") not a functional problem